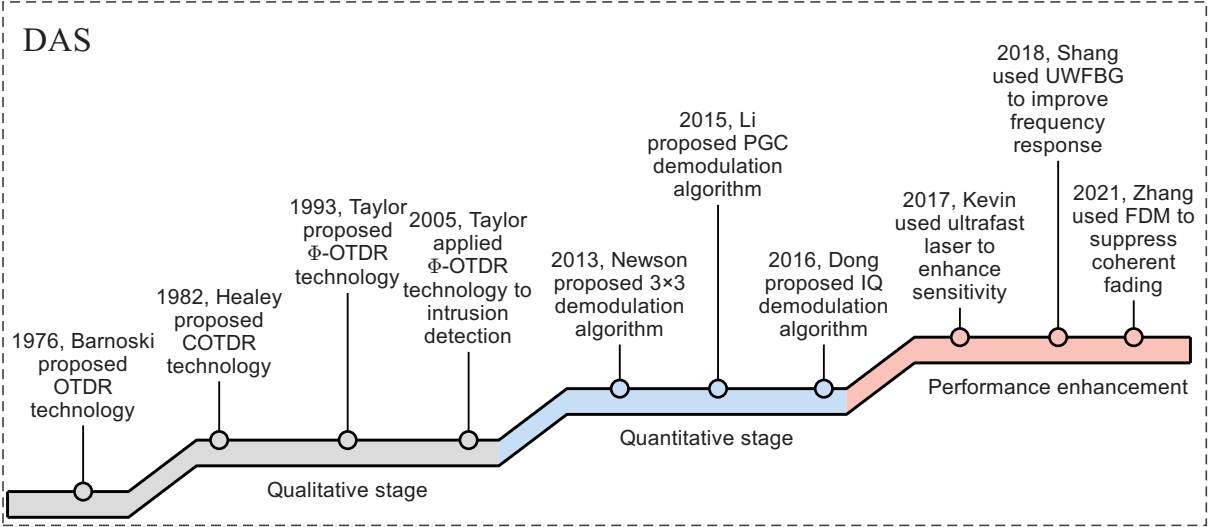
DAS
1976, Barnoski proposed OTDR technology
1982, Healey proposed COTDR technology
1993, Taylor proposed Φ-OTDR technology
2005, Taylor applied Φ-OTDR technology to intrusion detection
2013, Newson proposed 3×3 demodulation algorithm
2015, Li proposed PGC demodulation algorithm
2016, Dong proposed IQ demodulation algorithm
2017, Kevin used ultrafast laser to enhance sensitivity
2018, Shang used UWFBG to improve frequency response
2021, Zhang used FDM to suppress coherent fading
Qualitative stage
Quantitative stage
Performance enhancement